

§ 176. The real field H_m contained in Ω_m .

Every number of the field Ω_m is transformed by the substitution (r, r^{-1}) into the conjugate-imaginary number which is also obtained by the interchange $(i, -i)$. The product of two such conjugate-imaginary numbers is the square of the absolute value of either of them.

A number in Ω_m is real if it remains unchanged under the interchange (r, r^{-1}) , and only under this condition. Every such number can be expressed rationally by the two-term period $r + r^{-1}$, and therefore belongs to a real field of degree $\frac{1}{2}\varphi(m)$ which is a divisor of Ω_m . This field we shall call the real field contained in Ω_m and denote it by H_m (although other real fields are also contained in Ω_m , namely all divisors of H_m).

The $\frac{1}{2}\varphi(m) = \frac{1}{2}\mu$ numbers

$$1, \quad r + r^{-1}, \quad r^2 + r^{-2}, \dots, \quad r^{\frac{1}{2}\mu-1} + r^{-\frac{1}{2}\mu+1} \quad (1)$$

form a basis of the algebraic integers of the field H_m . For by § 170, (7),

$$\begin{aligned} \omega = & x_0 + x_1 r + x_2 r^2 + \dots + x_{\frac{1}{2}\mu-1} r^{\frac{1}{2}\mu-1} + x_{\frac{1}{2}\mu} r^{\frac{1}{2}\mu} \\ & + x_{-1} r^{-1} + x_{-2} r^{-2} + \dots + x_{-\frac{1}{2}\mu+1} r^{-\frac{1}{2}\mu+1} \end{aligned} \quad (2)$$

is an algebraic integer of the field Ω_m if and only if the coefficients $x_0, x_1, \dots$ are rational integers.

The condition that this number should belong to the field H_m is obtained by carrying out the substitution (r, r^{-1}) and setting the difference equal to zero. If one then also divides by $r - r^{-1}$, one obtains

$$\begin{aligned} 0 = & (x_1 - x_{-1}) + (x_2 - x_{-2})(r + r^{-1}) + \dots \\ & + (x_{\frac{1}{2}\mu-1} - x_{-\frac{1}{2}\mu+1}) \frac{r^{\frac{1}{2}\mu-1} - r^{-\frac{1}{2}\mu+1}}{r - r^{-1}} + x_{\frac{1}{2}\mu} \frac{r^{\frac{1}{2}\mu} - r^{-\frac{1}{2}\mu}}{r - r^{-1}}. \end{aligned}$$

The divisions can be carried out here; and if one then multiplies by $r^{\frac{1}{2}\mu-1}$, there results for r a rational equation of degree lower than the μ th, which can be satisfied only if all its coefficients vanish. This leads to the equations

$$x_{\frac{1}{2}\mu} = 0, \quad x_{\frac{1}{2}\mu-1} = x_{-\frac{1}{2}\mu+1}, \dots, \quad x_1 = x_{-1},$$

and hence every algebraic integer ω of the field H_m is contained in the form

$$\omega = x_0 + x_1(r + r^{-1}) + x_2(r^2 + r^{-2}) + \dots + x_{\frac{1}{2}\mu-1}(r^{\frac{1}{2}\mu-1} + r^{-\frac{1}{2}\mu+1}).$$

Instead of the basis (1), if one sets

$$\rho = r + r^{-1},$$

one may also choose the powers of ρ :

$$1, \quad \rho, \quad \rho^2, \dots, \quad \rho^{\frac{1}{2}\mu-1} \quad (3)$$

as a basis of the algebraic integers of H_m . For the quantities (3) can be expressed integrally in terms of the quantities (1), and conversely.

The number ρ is the root of an irreducible equation of degree $\frac{1}{2}\mu$, which we shall denote by $\psi(\rho) = 0$. If $f(r) = 0$ is the equation for r (§ 169), then the identical relation

$$f(x) = x^{\frac{1}{2}\mu} \psi\left(x + \frac{1}{x}\right) \quad (4)$$

holds; differentiating gives

$$f'(r) = r^{\frac{1}{2}\mu}\psi'(\rho)(1 - r^{-2}). \quad (5)$$

From this we can form the fundamental number Δ_1 of the field H_m , which, since H_m contains only real numbers, must be positive. If we denote by N_1 the norm with respect to H_m , this fundamental number is

$$\Delta_1 = \pm N_1\psi'(\rho).$$

But if N is the norm formed with respect to Ω_m , then

$$N\psi'(\rho) = [N_1\psi'(\rho)]^2 = \Delta_1^2,$$

and therefore formula (5), with regard to § 169, gives

$$\Delta = N(1 - r^{-2})\Delta_1^2.$$

By § 169, (7) and (10),

$$\begin{aligned} N(1 - r^{-2}) &= q && \text{for odd } q, \\ &= 4 && \text{for } q = 2, \\ \pm\Delta &= q\Delta_1^2 && \text{for odd } q, \\ &= 4\Delta_1^2 && \text{for } q = 2. \end{aligned} \quad (6)$$

If one substitutes therein

$$\Delta = \pm q^{q^{\kappa-1}[\kappa(q-1)-1]},$$

it follows that

$$\begin{aligned} \Delta_1 &= q^{q^{\kappa-1}\frac{q-1}{2} - \frac{q^{\kappa-1}+1}{2}} && \text{for odd } q, \\ &= 2^{(\kappa-1)2^{\kappa-2}-1} && \text{for } q = 2, \end{aligned} \quad (7)$$

so that Δ_1 is always a power of q .

§ 177. The prime ideals in the field H_m .

We now have to examine the question of the relation between the prime ideals of the field H_m and those of the field Ω_m .

If $\mathfrak{P}$ denotes any prime ideal in H_m of degree f_1 , then $\mathfrak{P}$ must be a divisor of a natural prime number p , and therefore (by § 171) $\mathfrak{P}$ can be divisible by the second or a higher power of a prime factor $\mathfrak{p}$ in Ω_m only if $p = q$.

If, however, $\mathfrak{P}$ is a divisor of q , then it must be a power of $\sigma = 1 - r$. Put, for instance,

$$\mathfrak{P} = \sigma^\lambda.$$

Taking the norm of this with respect to Ω_m , which is the square of the norm with respect to H_m , one obtains (by § 169)

$$q^{2f_1} = q^\lambda,$$

and hence $f_1 = \frac{1}{2}\lambda$. Since f_1 is an integer, λ must be at least equal to 2. But λ also cannot be greater than 2, since σ^2 is associated with the algebraic integer $(1 - r)(1 - r^{-1})$ contained in H_m , and therefore σ^2 must be divisible by $\mathfrak{P}$. Consequently $f_1 = 1$, and we obtain the theorem:

1. The prime number q is the $\frac{1}{2}\mu$ th power of a prime number of first degree existing in H_m .

Now let p be different from q , and let $\mathfrak{p}$ be a prime factor of $\mathfrak{P}$ in the field Ω_m , of degree f , which is carried by the substitution (r, r^{-1}) into $\mathfrak{p}'$. Then $\mathfrak{P}$ is divisible both by $\mathfrak{p}$ and by $\mathfrak{p}'$; on the other hand, $\mathfrak{p}\mathfrak{p}'$ is contained in H_m and is therefore divisible by $\mathfrak{P}$. Thus we have to distinguish whether $\mathfrak{p}$ and $\mathfrak{p}'$ are distinct or not.

If $\mathfrak{p}$ is distinct from $\mathfrak{p}'$, then $\mathfrak{P}$ is divisible by $\mathfrak{p}\mathfrak{p}'$, and it follows that $\mathfrak{P} = \mathfrak{p}\mathfrak{p}'$; taking norms gives $f_1 = f$.

If, however, $\mathfrak{p} = \mathfrak{p}'$, then $\mathfrak{P} = \mathfrak{p}$, and taking norms gives

$$2f_1 = f.$$

Which of these two cases occurs depends, therefore, on whether the prime ideal $\mathfrak{p}$ admits the substitution (r, r^{-1}) or not, that is, whether (r, r^{-1}) is contained in the group to which the prime ideal belongs or not. By § 172 this difference is determined by whether there is some exponent h for which the congruence

$$p^h \equiv -1 \pmod{m}$$

is satisfied, or whether there is no such exponent.

We summarize the result as follows:

2. The natural prime numbers p different from q split into two kinds, p_1, p_2 .

The first kind p_1 is characterized by the condition that for some exponent h

$$p_1^h \equiv -1 \pmod{m},$$

and these prime numbers, if they belong to the exponent f modulo m and if $\mu = ef$, split in the field H_m into e prime factors of degree $\frac{1}{2}f$. Their prime factors are not further decomposable in Ω_m either.

The prime numbers of the second kind p_2 are determined by the condition that none of their powers is congruent to -1 modulo m ; they split in the field H_m into $\frac{1}{2}e$ prime factors of degree f . Their prime factors are still decomposable in Ω_m into two factors each.

For prime numbers of the first kind, f must certainly be even. If m is odd, it is also sufficient, in order that p be a prime number of the first kind, that it belong to an even exponent; for then

$$(p^{\frac{1}{2}f} - 1)(p^{\frac{1}{2}f} + 1) \equiv 0 \pmod{m}.$$

The first of these factors, $p^{\frac{1}{2}f} - 1$, is not divisible by m , since otherwise p would belong to the exponent $\frac{1}{2}f$. Consequently $p^{\frac{1}{2}f} + 1$ must be divisible by q . But here both factors $p^{\frac{1}{2}f} - 1$ and $p^{\frac{1}{2}f} + 1$ cannot be divisible by q , since otherwise their difference, which is equal to 2, would also be divisible by q . Hence $p^{\frac{1}{2}f} + 1$ is divisible by m .

If, however, m is a power of 2, then the congruence

$$p^h \equiv -1 \pmod{m}$$

is possible only if $p \equiv -1 \pmod{m}$. For every even power of an odd number is $\equiv +1 \pmod{8}$ and therefore cannot be $\equiv -1 \pmod{m}$. But if h is odd, then

$$\frac{p^h + 1}{p + 1} = p^{h-1} - p^{h-2} + \dots + 1$$

is itself odd, and consequently $p^h + 1$ is divisible by no higher power of 2 than $p + 1$. Thus we may add to Theorem 2 the following more precise determination:

3. If m is odd, then the prime numbers p_1 belong to an even exponent, and the prime numbers p_2 to an odd exponent.

If m is even, then the prime numbers of the form $km - 1$ belong to the first kind, and all other odd prime numbers to the second kind.

§ 178. The units of the field H_m .

The units belonging to the field H_m are of special importance for the applications. The units of the field Ω_m may also be reduced to them, according to the following theorem:

1. Every unit of the field Ω_m is the product of a root of unity and a real unit of the field H_m .

Denote any unit of the field Ω_m by $\mathcal{E}(r)$. Then $\mathcal{E}(r^{-1})$ is the conjugate-imaginary value, which likewise represents a unit, and the quotient $\mathcal{E}(r) : \mathcal{E}(r^{-1})$ is again a unit whose absolute value

$$\sqrt{\frac{\mathcal{E}(r)}{\mathcal{E}(r^{-1})} \frac{\mathcal{E}(r^{-1})}{\mathcal{E}(r)}} \quad (1)$$

is equal to 1. Consequently this quotient is a root of unity and hence is $= \pm r^h$, where h is some integral exponent (§ 175, Theorems 1 and 2). We therefore have

$$\mathcal{E}(r) = \pm r^h \mathcal{E}(r^{-1}). \quad (2)$$

In the rest of the proof a small distinction has now to be made according as m is odd or even.

First suppose that m is odd. Then in (2) we may take the exponent h to be even, since, if necessary, we may replace it by $h + m$. Moreover, in formula (2) only the upper sign is admissible. For if the lower sign stood there, it would follow that

$$\mathcal{E}(r) \equiv \mathcal{E}(1) \equiv -\mathcal{E}(1), \quad 2\mathcal{E}(1) \equiv 0 \pmod{1-r}.$$

But by § 169, $1 - r$ is a divisor of q , and so is relatively prime to 2; therefore $\mathcal{E}(1)$, and consequently also $\mathcal{E}(r)$, is divisible by $(1 - r)$. This contradicts the supposition that $\mathcal{E}(r)$ is to be a unit.

Consequently (2) gives

$$r^{-\frac{1}{2}h} \mathcal{E}(r) = r^{\frac{1}{2}h} \mathcal{E}(r^{-1}),$$

whence it follows that $r^{-\frac{1}{2}h} \mathcal{E}(r)$ is a real unit. Denoting it by $e(r)$, we therefore have

$$\mathcal{E}(r) = r^{\frac{1}{2}h} e(r), \quad (3)$$

which proves Theorem 1 in this case.

If, secondly, m is a power of 2, then $\mu = \frac{1}{2}m$ and $r^\mu = -1$; and along with r , $-r$ is also an m th root of unity. If necessary replacing h by $h + \frac{1}{2}m$, we may assume the upper sign in (2) and write

$$\mathcal{E}(r) = r^h \mathcal{E}(r^{-1}). \quad (4)$$

The point is now to prove that h must be even.

If we represent $\mathcal{E}(r)$ in the basis $1, r, r^2, \dots, r^{\mu-1}$, then

$$\mathcal{E}(r) = \sum_{s=0}^{\mu-1} x_s r^s, \quad (5)$$

where the x_s are rational integers. But by (4) we then have

$$\sum_{s=0}^{\mu-1} x_s r^s = \sum_{s=1}^{\mu-1} x_s r^{h-s},$$

and from this it follows that $x_s = \pm x_{s'}$ whenever $s + s' \equiv h \pmod{\mu}$. If h is odd, then of two numbers so connected one is even and the other odd (since μ is a power of 2). Thus the expression (5) gives for $\mathcal{E}(r)$ an aggregate of terms of the form

$$x_s(r^s \pm r^{s'}).$$

But

$$r^s \pm r^{s'} = r^s(1 \pm r^{s'-s}) = r^s(1 \mp r^{\mu+s'-s})$$

is always divisible by $1 - r$, and from this it would follow that $\mathcal{E}(r)$ is divisible by $1 - r$, which is not a unit, contrary to the concept of a unit. Therefore the assumption that the exponent h in formula (4) is odd is inadmissible, and, as above, we may put

$$\mathcal{E}(r) = r^{\frac{1}{2}h} e(r),$$

where $e(r)$ is a real unit. Theorem 1 is thereby proved.

We shall single out a system of real units. By § 169 the μ quantities $1 - r^n$ are all associated with one another. Therefore the quotient

$$\frac{1 - r^n}{1 - r^{n'}},$$

where n, n' are two numbers relatively prime to m , is a unit. From this, however, setting

$$r = e^{\frac{2\pi i}{m}},$$

one derives the real unit

$$\frac{\sin \frac{n\pi}{m}}{\sin \frac{n'\pi}{m}} = \frac{r^{\frac{n}{2}} - r^{-\frac{n}{2}}}{r^{\frac{n'}{2}} - r^{-\frac{n'}{2}}} = r^{\frac{n'-n}{2}} \frac{1 - r^n}{1 - r^{n'}}. \quad (6)$$

If m is even, one may set $n' = n + \frac{m}{2}$, and obtains in this case the real units

$$\tau_n = \tan \frac{n\pi}{m}, \quad (7)$$

where n may denote any odd number.

Replacing n by $\frac{1}{2}m - n$ carries τ_n into its reciprocal value. If n runs through the sequence of numbers

$$1, 3, 5, \dots, \frac{m}{4} - 1,$$

then all the values τ_n obtained are positive proper fractions.